

## 0.1 Spacetime symmetries

1. For an infinitesimal boost in the  $[0i]$  plane, the spacetime coordinates transform as:

$$t \rightarrow t - \eta x^i \quad \text{and} \quad x^i \rightarrow x^i - \eta t,$$

therefore we get the Killing vector field corresponding to the boost:

$$k^\mu = \frac{\partial x^\mu}{\partial \eta} = (-x^i, -t\delta_i^\mu)^T \implies k = -x^i \partial_t - t \partial_{x^i}$$

Overall we obtain for the conserved current:

$$\begin{aligned} J_\mu^{[0i]} &= \frac{\partial \mathcal{L}}{\partial(\partial^\mu \phi)} k^\nu \delta_\nu \phi + \frac{\partial \mathcal{L}}{\partial(\partial^\mu \phi^*)} k^\nu \delta_\nu \phi^* - k_\mu \mathcal{L} \\ &= \partial_\mu \phi^* (-x^i \partial_t - t \partial_{x^i}) \phi + \partial_\mu \phi (-x^i \partial_t - t \partial_{x^i}) \phi^* - k_\mu \mathcal{L} \\ &= -\partial_\mu \phi^* (x^i \partial_t + t \partial_{x^i}) \phi - \partial_\mu \phi (x^i \partial_t + t \partial_{x^i}) \phi^* - (\eta_\mu^i t - \eta_\mu^0 x^i) \mathcal{L} \end{aligned}$$

2. Using the Klein-Gordon equation for each independent field

$$(\partial^\mu \partial_\mu + m^2) \phi = 0 \quad \text{and} \quad (\partial^\mu \partial_\mu + m^2) \phi^* = 0$$

we get

$$\begin{aligned} \partial^\mu J_\mu^{[0i]} &= -(\partial^\mu \partial_\mu \phi^*) (x^i \partial_t + t \partial_{x^i}) \phi - \partial_\mu \phi^* (\delta^{\mu i} \partial_t + \delta^{\mu t} \partial_{x^i}) \phi - \partial_\mu \phi^* (x^i \partial_t \\ &\quad + t \partial_{x^i}) \partial^\mu \phi - c.c. - (\eta_\mu^i \delta^{\mu t} - \eta_\mu^0 \delta^{\mu i}) \mathcal{L} \\ &\quad - (\eta_\mu^i t - \eta_\mu^0 x^i) [\partial_\nu \phi^* \partial^\mu \partial^\nu \phi + m^2 \phi^* \partial^\mu \phi] - c.c. \\ &= 0, \end{aligned}$$

where the following terms cancel:

- $(\eta_\mu^i \delta^{\mu t} - \eta_\mu^0 \delta^{\mu i}) = 0$  using the diagonality of the Minkowski metric
- $-(\partial^\mu \partial_\mu \phi^*) (x^i \partial_t + t \partial_{x^i}) \phi - (\eta_\mu^i t - \eta_\mu^0 x^i) m^2 \phi^* \partial^\mu \phi = 0$  using the EOM
- $-\partial_\mu \phi^* (x^i \partial_t + t \partial_{x^i}) \partial^\mu \phi - (\eta_\mu^i t - \eta_\mu^0 x^i) \partial_\nu \phi^* \partial^\mu \partial^\nu \phi = 0$
- the complex conjugate terms similar to the two above cancel each other as well
- $-\partial_\mu \phi^* (\delta^{\mu i} \partial_t + \delta^{\mu t} \partial_{x^i}) \phi - \partial_\mu \phi (\delta^{\mu i} \partial_t + \delta^{\mu t} \partial_{x^i}) \phi^*$  cancel as we can write  $\delta^{\mu i} = -\eta^{\mu i}$  and  $\delta^{\mu t} = \eta^{\mu t}$  and the cross terms cancel each other

3. We can write

$$\begin{aligned}\partial^\mu J_\mu^{(A)} &= (\partial^\mu T_{\mu\nu})k_{(A)}^\nu + T_{\mu\nu}(\partial^\mu k_{(A)}^\nu) \\ &= 0 + \frac{1}{2}T_{\mu\nu} \left[ \partial^\mu k_{(A)}^\nu + \partial^\nu k_{(A)}^\mu \right] + \frac{1}{2} \left[ \partial^\mu k_{(A)}^\nu - \partial^\nu k_{(A)}^\mu \right] \\ &= \frac{1}{2}T_{\mu\nu} \cdot 0 - 0 = 0,\end{aligned}$$

where we have used the conservation of the energy-stress tensor in the first step, the Killing equation and the symmetry of energy-stress tensor in  $(\mu\nu)$  in the second step.

In the (optional) part we have the Killing field for translations in direction  $(\nu)$ :

$$x^\mu \rightarrow x^\mu + \eta \delta_{(\nu)}^\mu \implies k_{(\nu)}^\mu = \delta_{(\nu)}^\mu \implies k = \delta_{(\nu)}^\mu \partial_\mu$$

and therefore we get:

$$J_{\mu(\nu)} = \partial_\mu \phi^* \delta_{(\nu)}^\kappa \partial_\kappa \phi + \partial_\mu \phi \delta_{(\nu)}^\kappa \partial_\kappa \phi^* - \delta_{\mu(\nu)} \mathcal{L} =: T_{\mu(\nu)}$$

Using this relation we obtain

$$\begin{aligned}J_\mu^{[0i]} &= T_{\mu\nu} k_{[0i]}^\nu = [\partial_\mu \phi^* \delta_\nu^\kappa \partial_\kappa \phi + \partial_\mu \phi \delta_\nu^\kappa \partial_\kappa \phi^* - \delta_{\mu\nu} \mathcal{L}] [-\delta_t^\nu x^i - \delta_i^\nu t] \\ &= -\partial_\mu \phi^* (x^i \partial_t + t \partial_{x^i}) \phi - \partial_\mu \phi (x^i \partial_t + t \partial_{x^i}) \phi^* + (\delta_\mu^i t + \delta_\mu^t x^i) \mathcal{L},\end{aligned}$$

which is the same expression as the one obtained in the first exercise.

4. For the conserved charge we have

$$\begin{aligned}Q^{[0i]} &= \int_{t=\text{const.}} d^3 \mathbf{x} J_0^{[0i]} \\ &= \int_{t=\text{const.}} d^3 \mathbf{x} [-\partial_t \phi^* (x^i \partial_t + t \partial_{x^i}) \phi - \partial_t \phi (x^i \partial_t + t \partial_{x^i}) \phi^* - (\eta_t^i t - \eta_t^t x^i) \mathcal{L}] \\ &= - \int_{t=\text{const.}} d^3 \mathbf{x} x^i [\partial_t \phi^* \partial_t \phi + \partial_t \phi \partial_t \phi^* - \mathcal{L}] - t \int_{t=\text{const.}} d^3 \mathbf{x} [\partial_t \phi^* \partial_i \phi + \partial_t \phi \partial_i \phi^*] \\ &= - \int_{t=\text{const.}} d^3 \mathbf{x} x^i \mathcal{H} - t \int_{t=\text{const.}} d^3 \mathbf{x} T_{0i}\end{aligned}$$

5. We can express the fields  $\phi$  and  $\phi^*$  using the creation and annihilation op-

erators for (anti)particles:

$$\begin{aligned}\phi(x) &= \int \tilde{d}\mathbf{k} \left[ e^{-ikx} \hat{a}_+(k) + e^{ikx} \hat{a}_-^\dagger(k) \right] \\ \phi^*(x) \equiv \phi^\dagger(x) &= \int \tilde{d}\mathbf{k} \left[ e^{-ikx} \hat{a}_-(k) + e^{ikx} \hat{a}_+^\dagger(k) \right]\end{aligned}$$

and therefore also their time derivatives:

$$\begin{aligned}\pi^\dagger(x) &= \partial_t \phi(x) = -i \int \tilde{d}\mathbf{k} E(\mathbf{k}) \left[ e^{-ikx} \hat{a}_+(k) - e^{ikx} \hat{a}_-^\dagger(k) \right] \\ \pi(x) &= \partial_t \phi^\dagger(x) = -i \int \tilde{d}\mathbf{k} E(\mathbf{k}) \left[ e^{-ikx} \hat{a}_-(k) - e^{ikx} \hat{a}_+^\dagger(k) \right]\end{aligned}$$

or in general:

$$\begin{aligned}\partial_\mu \phi(x) &= -i \int \tilde{d}\mathbf{k} k_\mu \left[ e^{-ikx} \hat{a}_+(k) - e^{ikx} \hat{a}_-^\dagger(k) \right] \\ \partial_\mu \phi^\dagger(x) &= -i \int \tilde{d}\mathbf{k} k_\mu \left[ e^{-ikx} \hat{a}_-(k) - e^{ikx} \hat{a}_+^\dagger(k) \right]\end{aligned}$$

Since the conserved charge is constant in time  $t$ , we can compute it at  $t = 0$ , therefore only the first term survives. Then we have

$$\begin{aligned}Q^{[0i]} &= - \int d^3\mathbf{x} x^i \mathcal{H} = - \int d^3\mathbf{x} x^i \left[ \partial_\mu \phi^* \partial^\mu \phi + m^2 \phi^* \phi \right] \\ &= \int \tilde{d}\mathbf{k} \tilde{d}\mathbf{q} d^3\mathbf{x} x^i \left[ k_\mu q^\mu \left( e^{i(\mathbf{k}+\mathbf{q})\cdot\mathbf{x}} \hat{a}_-(k) \hat{a}_+(q) - e^{i(\mathbf{k}-\mathbf{q})\cdot\mathbf{x}} \hat{a}_-(k) \hat{a}_-^\dagger(q) \right. \right. \\ &\quad \left. \left. - e^{-i(\mathbf{k}-\mathbf{q})\cdot\mathbf{x}} \hat{a}_+^\dagger(k) \hat{a}_+(q) + e^{-i(\mathbf{k}+\mathbf{q})\cdot\mathbf{x}} \hat{a}_+^\dagger(k) \hat{a}_-^\dagger(q) \right) - m^2 \left( e^{i(\mathbf{k}+\mathbf{q})\cdot\mathbf{x}} \hat{a}_-(k) \hat{a}_+(q) \right. \right. \\ &\quad \left. \left. + e^{i(\mathbf{k}-\mathbf{q})\cdot\mathbf{x}} \hat{a}_-(k) \hat{a}_-^\dagger(q) + e^{-i(\mathbf{k}-\mathbf{q})\cdot\mathbf{x}} \hat{a}_+^\dagger(k) \hat{a}_+(q) + e^{-i(\mathbf{k}+\mathbf{q})\cdot\mathbf{x}} \hat{a}_+^\dagger(k) \hat{a}_-^\dagger(q) \right) \right]\end{aligned}$$

Now we can use the fact that

$$\int d^3\mathbf{x} x^i e^{i\mathbf{p}\cdot\mathbf{x}} = -i(2\pi)^3 \frac{\partial}{\partial p_i} \delta^{(3)}(\mathbf{p})$$

and also

$$-i(2\pi)^3 \int \tilde{d}\mathbf{p} f(\mathbf{p}) \frac{\partial}{\partial p_i} \delta^{(3)}(\mathbf{p}) = i(2\pi)^3 \int \tilde{d}\mathbf{p} \frac{\partial f(\mathbf{p})}{\partial p_i} \delta^{(3)}(\mathbf{p}) = \frac{i}{2E(\mathbf{p})} \frac{\partial f(\mathbf{p})}{\partial p_i} \Big|_{\mathbf{p}=\mathbf{0}}$$

Therefore we get

$$\begin{aligned} Q^{[0i]} = & \frac{i}{2} \int \frac{\tilde{d}\mathbf{k}}{E(\mathbf{k})} \left[ +\tilde{k}_\mu \hat{a}_-(\tilde{k}) \frac{\partial}{\partial k_i} (k^\mu \hat{a}_+(k)) + k_\mu \hat{a}_-(k) \frac{\partial}{\partial k_i} (k^\mu \hat{a}_-^\dagger(k)) \right. \\ & \left. - k_\mu \hat{a}_+^\dagger(k) \frac{\partial}{\partial k_i} (k^\mu \hat{a}_+(k)) - \tilde{k}_\mu \hat{a}_+^\dagger(\tilde{k}) \frac{\partial}{\partial k_i} (k^\mu \hat{a}_-^\dagger(k)) \right] \\ & - \frac{i}{2} m^2 \int \frac{\tilde{d}\mathbf{k}}{E(\mathbf{k})} \left[ +\hat{a}_-(\tilde{k}) \frac{\partial}{\partial k_i} \hat{a}_+(k) - \hat{a}_-(k) \frac{\partial}{\partial k_i} \hat{a}_-^\dagger(k) \right. \\ & \left. + \hat{a}_+^\dagger(k) \frac{\partial}{\partial k_i} \hat{a}_+(k) - \hat{a}_+^\dagger(\tilde{k}) \frac{\partial}{\partial k_i} \hat{a}_-^\dagger(k) \right] \end{aligned}$$

Now we have to apply the Leibnitz rule

$$\begin{aligned} \frac{\partial}{\partial k_i} (k^j \hat{a}(k)) &= \frac{\partial k^j}{\partial k_i} \hat{a}(k) + k^j \partial^i \hat{a}(k) = \delta^{ji} \hat{a}(k) + k^j \partial^i \hat{a}(k), \\ \frac{\partial}{\partial k_i} (k^0 \hat{a}(k)) &= \frac{\partial E(\mathbf{k})}{\partial k_i} \hat{a}(k) + E(\mathbf{k}) \partial^i \hat{a}(k) = \frac{k^i}{E(\mathbf{k})} \hat{a}(k) + E(\mathbf{k}) \partial^i \hat{a}(k), \end{aligned}$$

where these terms are then multiplied by  $\frac{1}{E(\mathbf{k})} k_0 = \frac{1}{E(\mathbf{k})} E(\mathbf{k}) = 1$  for the time component and  $\frac{1}{E(\mathbf{k})} k_j = -\frac{1}{E(\mathbf{k})} k^j$  for the space components. We can also rewrite

$$\frac{m^2}{E(\mathbf{k})} = E(\mathbf{k}) - \frac{\mathbf{k}^2}{E(\mathbf{k})}.$$

Using this relation after applying the Leibnitz rule, we can rewrite our integral in terms of three sums with prefactors  $E(\mathbf{k})$ ,  $\frac{\mathbf{k}^2}{E(\mathbf{k})}$  with derivatives of creation / annihilation operators and  $\frac{k^i}{E(\mathbf{k})}$  without derivatives. Each of these sums will contain eight elements, with each operator pair appearing twice. The operator pairs with one operator depending on  $\tilde{k}$  appear always with a different sign and cancel each other out, while the operator pairs depending only on  $k$  have different sign only in the  $\frac{\mathbf{k}^2}{E(\mathbf{k})}$  sum, where they cancel, however they have the same sign in the other two sums, where they add up. This way we obtain the final relation:

$$\begin{aligned} Q^{[0i]} = & i \int \tilde{d}k \left[ E(\mathbf{k}) \left( \hat{a}_-(k) \partial^i \hat{a}_-^\dagger(k) - \hat{a}_+^\dagger(k) \partial^i \hat{a}_+(k) \right) \right. \\ & \left. + \frac{k^i}{E(\mathbf{k})} \left( \hat{a}_-(k) \hat{a}_-^\dagger(k) - \hat{a}_+^\dagger(k) \hat{a}_+(k) \right) \right] \end{aligned}$$

## 0.2 Target space symmetries

[1.] We can see that because the matter matrix of fields is diagonal, any rotation of the fields does not change the Lagrangian density, i.e.:

$$\phi \rightarrow R\phi, \quad \text{where} \quad RR^T = R^T R = \mathbb{1}, R \in \mathbb{R}^{N \times N}$$

therefore the group of the global symmetry is  $O(N)$ . Here, I am not certain if the continuous global symmetry we are searching means we should only look at its subgroup  $SO(N)$ , as  $O(N)$  also contains elements corresponding to discontinuous symmetries such as reflections?

[2.] For the generators of this group the following relation must hold

$$M + M^T = 0,$$

i.e. we have the same number of generators as is the number of  $N \times N$  antisymmetric real matrices:

$$\dim SO(N) = \frac{1}{2}N(N-1).$$

The infinitesimal transformations of field can be expressed using the Kronecker delta. Rotations can be performed in planes  $(ij)$ , where  $i, j \in \{1, \dots, N\}, i \neq j$ . This fact corresponds to the dimension of the group, as there is  $\binom{N}{2} = \frac{1}{2}N(N-1)$  ways to pick two distinct indices  $(ij)$  without repetition.

$$\delta_{(ij)}\phi^a = \delta_{(i)}^a\phi^{(j)} - \delta_{(j)}^a\phi^{(i)} = \left[ \phi^{(j)} \frac{\partial}{\partial \phi^{(i)}} - \phi^{(i)} \frac{\partial}{\partial \phi^{(j)}} \right] \phi^a$$

For the conserving currents we then have

$$\begin{aligned} J_{(ij)}^\mu &= \sum_{a=1}^N \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^a)} \delta_{(ij)}\phi^a + 0 \\ &= \sum_{a=1}^N \partial^\mu \phi^a \left[ \phi^{(j)} \frac{\partial}{\partial \phi^{(i)}} - \phi^{(i)} \frac{\partial}{\partial \phi^{(j)}} \right] \phi^a \\ &= \partial^\mu \phi^{(i)} \phi^{(j)} - \partial^\mu \phi^{(j)} \phi^{(i)} \end{aligned}$$

[3.] For the conserved charges we can write

$$Q_{(ij)} = - \int_{t=\text{const.}} d^3x J_{(ij)}^0 = - \int_{t=\text{const.}} d^3x [\pi^{(i)}\phi^{(j)} - \pi^{(j)}\phi^{(i)}]$$

We can now express the field using annihilation and creation operators

$$\begin{aligned}\phi^a(x) &= \int \tilde{d}k [e^{-ikx} \hat{a}_a(k) + e^{ikx} \hat{a}_a^\dagger(k)] \\ \pi^a(x) &= -i \int \tilde{d}q E(\mathbf{q}) [e^{-iqx} \hat{a}_a(q) - e^{iqx} \hat{a}_a^\dagger(q)]\end{aligned}$$

By substituting these relations back into the formula for conserved charges and taking  $t = 0$ , we obtain

$$\begin{aligned}Q_{(ij)} &= i \int \tilde{d}q \tilde{d}k d^3x E(\mathbf{q}) \left[ \left( e^{i\mathbf{q}\cdot\mathbf{x}} \hat{a}_{(i)}(q) - e^{-i\mathbf{q}\cdot\mathbf{x}} \hat{a}_{(i)}^\dagger(q) \right) \left( e^{i\mathbf{k}\cdot\mathbf{x}} \hat{a}_{(j)}(k) + e^{-i\mathbf{k}\cdot\mathbf{x}} \hat{a}_{(j)}^\dagger(k) \right) \right. \\ &\quad \left. - \left( e^{i\mathbf{q}\cdot\mathbf{x}} \hat{a}_{(j)}(q) - e^{-i\mathbf{q}\cdot\mathbf{x}} \hat{a}_{(j)}^\dagger(q) \right) \left( e^{i\mathbf{k}\cdot\mathbf{x}} \hat{a}_{(i)}(k) + e^{-i\mathbf{k}\cdot\mathbf{x}} \hat{a}_{(i)}^\dagger(k) \right) \right] \\ &= \frac{i}{2} \int \tilde{d}k \left[ \hat{a}_{(i)}(\tilde{k}) \hat{a}_{(j)}(k) + \hat{a}_{(i)}(k) \hat{a}_{(j)}^\dagger(k) - \hat{a}_{(i)}^\dagger(k) \hat{a}_{(j)}(k) - \hat{a}_{(i)}^\dagger(\tilde{k}) \hat{a}_{(j)}^\dagger(k) \right. \\ &\quad \left. - \hat{a}_{(j)}(\tilde{k}) \hat{a}_{(i)}(k) - \hat{a}_{(j)}(k) \hat{a}_{(i)}^\dagger(k) + \hat{a}_{(j)}^\dagger(k) \hat{a}_{(i)}(k) + \hat{a}_{(j)}^\dagger(\tilde{k}) \hat{a}_{(i)}^\dagger(k) \right] \\ &= i \int \tilde{d}k \left[ \hat{a}_{(i)}^\dagger(k) \hat{a}_{(j)}(k) - \hat{a}_{(j)}^\dagger(k) \hat{a}_{(i)}(k) \right]\end{aligned}$$